

Chapter 20 — Constitutional Firewall

Field	Value
Status	Revised after whole-book constitutional and cross-chapter audit
Version	0.3
Scope	Mediate every exchange with an External Trust Domain so collaboration expands capability without transferring constitutional authority or uncontrolled data.
Constitutional basis	Decisions 26–27, 32, 36, 39–40, 48–50
Owner review	No unresolved Owner decision after whole-book audit

1. Purpose

Mediate every exchange with an External Trust Domain so collaboration expands capability without transferring constitutional authority or uncontrolled data.

2. Authoritative Responsibilities

Component	Authoritative responsibility
Treaty Manager	Treaty identity, scope, lifecycle, renewal, revocation and evidence
Constitutional Firewall	Ingress/egress validation, redaction, authorization and audit
External Domain Registry	Domain identity, authentication, capabilities, trust and active Treaties
Cross-Domain Monitor	Exchange health, policy drift, incidents and revocation signals

3. Core State and Records

Object	Required content
Treaty	Parties, purpose, capabilities, data scope, authentication, privacy, duration, renewal, revocation and audit.
Exchange Envelope	Sender/recipient, Treaty, intent, classification, provenance, authorization, replay protection and expiry.
Redaction Decision	Requested fields, permitted fields, removals, rationale and policy version.
Domain Incident	Violation, affected exchanges, containment, evidence, notification and recovery.

4. Runtime Workflow

1. Authenticate the external domain and resolve an active Treaty for the requested purpose.
2. Validate that the capability class was Owner-approved and the specific provider/domain is permitted.
3. Evaluate sender authority, data classification, purpose, requested fields and recipient rights.
4. Minimize/redact egress data and verify ingress provenance, integrity, schema and allowed meaning.
5. Bind the exchange to a transaction or communication object and record audit metadata.
6. Revoke, expire or suspend exchanges immediately when Treaty, identity or integrity conditions fail.

5. Interfaces and Contracts

All commands, queries and events are typed, versioned, authenticated and correlated. Commands request mutation from the authoritative owner; queries declare consistency and privacy scope; immutable events record completed facts with identity, causation, authorization, provenance and integrity metadata.

6. Failure and Recovery

- No Treaty or expired Treaty: deny the exchange; local capability availability is irrelevant.
- Provenance failure: quarantine incoming evidence and prevent state mutation.
- Overbroad data request: redact or deny; never infer consent from prior collaboration.
- External compromise: revoke credentials/Treaty access, preserve evidence and contain affected transactions.

7. Constitutional Guarantees

- HAL remains constitutionally sovereign in every federation.
- Cooperation expands capability and never transfers authority.
- Every cross-domain exchange is authenticated, authorized, minimized, provenance-checked and auditable.
- Treaty approval and capability-class approval are distinct protected decisions.

8. Prohibited Behaviors

- No component may silently broaden authority, reinterpret Owner intent, erase dissent, or treat a derived projection as authoritative state.
- No degraded dependency may silently change policy, identity, trust, authentication, verification or evidence requirements.
- No provider, model, node, Presence, secret, relationship or network location receives constitutional authority by implication.

9. Observability and Verification

The subsystem records correlation, identity, policy/authority context, state versions, decisions, limitations, failures, recovery and outcomes. Verification is risk-scaled and produces durable evidence. Status reporting is derived from governed state rather than model assertion.

10. Source Alignment and Review

This chapter directly implements Decisions 26–27, 32, 36, 39–40, 48–50. The v0.2 rewrite restores the source-specific objects, workflows and guarantees missing from v0.1. Final disposition is recorded in the separate source-alignment and constitutional-audit reports.

Whole-Book Audit Resolution

The following requirements supersede any ambiguous earlier wording in this chapter and are incorporated into this edition:

- Chapter 21's Treaty Manager is the sole owner of Treaty proposal, approval, lifecycle, and history.
- The Constitutional Firewall owns exchange enforcement, redaction, ingress/egress validation, and cross-domain incident records and consumes signed Treaty state.